WINSTON TRINH

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EDUCATION

University of Southern California, Viterbi School of Engineering Los Angeles, CA
Bachelor of Science - Computer Science December 2024
Coursework: Software Engineering, Operating Systems, Algorithms, Object Orientation Programming and Design, Artificial Intelligence, Probability, Discrete Mathematics, Professional C++, Internetworking, Embedded Systems.

EXPERIENCE

AT&T Seattle, WA
Sr. Specialist Member of Technical Staff October 2026-Present
- AI Workflow Automation

Atlanta, GA
TDP Network Engineer July 2025-Present
- Identified **cell configuration optimization opportunities** by building an **end-to-end ETL pipeline** analyzing cell-level **LTE and 5G NR uplink KPIs** to diagnose **network conditions** and **performance bottlenecks**.
- Engineered **Python-based automated testing workflows** that reduced manual QA defects by **70%**, cut validation time by **30%**, and increased **deployment capacity by 25%**, enabling releases with more features for a wider range of stakeholder groups.
- Prototyped and designed a **Jira analytics dashboard** tracking developer velocity, defect trends, and release metrics, providing new visibility into productivity and informing data-driven process improvements.

USC Sol Price School of Public Policy Los Angeles, CA
Research Assistant July 2024-December 2024
- Process and analyze large-scale datasets exceeding 100 GB using Python and Polars to reduce data processing time.
- Develop and implement predictive models using scikit-learn and linearmodels to support nonprofit budget forecasting analysis.

CS@SC Los Angeles, CA
Teaching Assistant for USC Viterbi School of Engineering June 2022-July 2024
- Lead **weekly classes of up to 15 students** spanning K-12th grade, teaching various computer science topics ranging from Scratch to Java/Python.
- Design **lesson plans for 200+ students** with interactive programming problems and homework to support practical learning.
- Advise in development of Game Design using Unity with C# and Level 1 Python by recording 3 educational videos.

PROJECTS

Supabase Rate Limiting System Beaverton, OR
PostgreSQL, PL/pgSQL, Bash, Supabase March 2025
- Built a custom rate limiting backend using Supabase and PL/pgSQL, enabling precise control over API usage across different routes and users.
- Built RPC functions to enforce dynamic request quotas and manage system-level traffic with minimal latency.
- **Automated testing** with a **Bash script simulating real-time API traffic**, using realistic seed data to validate behavior under edge cases and production-like conditions.

Course Schedule Checker Beaverton, OR
Python, AWS Lambda, SNS, DynamoDB, IAM, cron February 2025
- Reverse-engineered UCLA's course scheduling site by analyzing browser network activity to replicate undocumented API requests.
- Deployed cURL-based polling logic in AWS Lambda, storing results in DynamoDB and sending seat availability alerts via SNS alerts.

SKILLS

Languages: C/C++, Python, Bash, SQL, C#, Java, TypeScript, JavaScript, HTML/CSS, Swift, Dart.
Frameworks: Cucumber, Playwright, Selenium, React.js, Next.js, Spring Boot, Flask, Flutter, Tailwind.
Tools: Git, Docker, AWS EC2, AWS SNS, Linux, GitHub Actions (CI/CD), Jira, Postman, Maven, Supabase, Firebase.
Databases: PostgreSQL, Snowflake, MySQL, DynamoDB, Firestore.
Concepts: Agile/Scrum, User Acceptance Testing, Unit Testing, A/B Testing, SAFe Agile, REST APIs, Algorithms, System Design.